APPLIED SCIENCES AND ENGINEERING

Hybrid locomotion at the insect scale: Combined flying and jumping for enhanced efficiency and versatility

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Insect-scale robots face two major locomotive challenges: constrained energetics and large obstacles that far exceed their size. Terrestrial locomotion is efficient yet mostly limited to flat surfaces. In contrast, flight is versatile for overcoming obstacles but requires high power to stay aloft. Here, we present a hopping design that combines a subgram flapping-wing robot with a telescopic leg. Our robot can hop continuously while controlling jump height and frequency in the range of 1.5 to 20 centimeters and 2 to 8.4 hertz. The robot can follow positional set points, overcome tall obstacles, and traverse challenging surfaces. It can also hop on a dynamically rotating plane, recover from strong collisions, and perform somersaults. Compared to flight, this design reduces power consumption by 64 percent and increases payload by 10 times. Although the robot relies on offboard power and control, the substantial payload and efficiency improvement open opportunities for future study on autonomous locomotion.

INTRODUCTION

Owing to their small size, insects need to frequently evade, traverse, or interact with objects orders of magnitude larger than themselves. Many insect species, such as ants, honeybees, and springtails, have developed remarkable locomotive capabilities to overcome large vertical obstacles. Depending on terrain roughness, obstacle size, and the presence of predators or preys, insects may crawl, climb, jump, glide, or fly as they move through unstructured environments (*1*, *2*). This multimodal strategy offers flexibility for insects to balance trade-offs between locomotion versatility and efficiency—the ability to overcome obstacles in their paths and the energetic cost corresponding to the locomotor mode (*3*). The energetic efficiency of an insect or robot can be measured by different metrics on the basis of locomotor mode and trajectory type. For instance, the hovering efficiency of aerial systems is best described by lift-to-power ratio that characterizes maximum lift force production per unit power. For robots or animals that move from one location to another, the metric cost of transport (COT) characterizes energetic cost per unit distance.

Addressing the trade-off relationship between locomotion efficiency and versatility is critical for designing insect-scale robots that face similar challenges. Most microrobots at the gram scale adopt a legged design (*4*–*8*), and they have demonstrated various gaits, speeds, and efficiencies. Although crawling and running motions are typically energetically efficient, small robots struggle when they confront rough terrains, large obstacles, or compliant and slippery surfaces such as leaves and flowers. A few legged microrobots can climb vertical surfaces (*9*, *10*) via electrostatic adhesion, but they are confined to smooth surfaces and suffer low speed and reliability. In contrast, flight demonstrates high locomotion versatility because small robots can easily fly over large and irregular objects. Aerial insects can evade ground predators and travel long distances to search for food, yet they must afford substantially higher energetic cost to stay aloft. From the microrobotic perspective, this energetic challenge exacerbates because of limited payload and a lack of energy sources. Existing subgram aerial robots cannot carry onboard energy sources (*11*–*13*), which severely limits their applications.

Jumping is ubiquitous among insects because it offers high locomotion versatility with moderate efficiency. Most jumping insects leverage latch-mediated spring actuation (*14*) principles to store and impulsively release energy—leading to high jumps that exceed 0.7 m or 115 times their body length (*15*). Subgram jumping robots (*16*, *17*) adopt a similar design where they can reach up to 0.64-m jump height. However, unlike insects, small robots lack the ability to control attitude and landing position while they are aloft, and they cannot reorient themselves and reload the jumping mechanism after landing. Achieving the ability to perform consecutive jumps can substantially improve microrobot mobility. Hopping, or continuous jumping, has been achieved in mesoscale (35 to 100 g) robots (*18*–*20*) to demonstrate efficient and versatile locomotion. Unlike single-jump robots (*21*, *22*) that slowly store energy in a spring-latched system, hopping robots (*18*, *23*) need to quickly store energy in series-elastic mechanisms during the aerial phase and release energy during the stance phase. Most hopping designs are based on the spring-loaded inverted pendulum model (*24*), and the mechanisms are realized on the basis of a combination of multiple linkages and springs (*18*–*20*).

However, these mesoscale designs are difficult to implement in insect-scale robots owing to scaling laws, fabrication, and control challenges. As robot size shrinks, robot dynamics become substantially faster, requiring rapid actuation and control in the range of tens of milliseconds. For a subgram robot to hop, the duration of ground contact becomes shorter than 20 ms, exceeding the control bandwidth of microscale actuators. The short ground impact can also cause large body torques and induce fast rotations, which require another set of actuators and mechanisms to stabilize robot's attitude in the aerial phase. In addition, it becomes increasingly difficult to construct nonlinear springs, flexures, and linkages at the

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submillimeter scale. Owing to these challenges, existing insect-scale robots have not achieved hopping despite the potential advantages of versatility and efficiency. At the tens-of-gram scale, our prior work presented a different hopping design by adding a passive telescopic leg to a quadcopter, resulting in a 35-g Hopcopter (*20*). This Hopcopter demonstrated better controllability and efficiency over other existing designs because it could inject energy and exert stronger attitude control in the aerial phase. Applying this design to insect-scale systems is advantageous because it reduces mechanical complexity. However, new challenges arise because of diminishing robot inertia and substantially faster system dynamics that require fast planning and attitude control. Specifically, high-bandwidth actuators and computationally efficient controllers are needed to replace electromagnetic motors and model-based collision planners (*20*).

In this work, we demonstrate efficient, versatile, and robust hopping in subgram robots by augmenting a micro–aerial vehicle with a passive elastic leg (Fig. 1, A to C, and movie S1). Compared to flight, this hopping design reduces energetic cost by 64% and increases the robot's payload by over 10 times. The robot can precisely control its jump height, frequency, and landing positions to track set points and leap over large obstacles. This hopping design is also adaptive to a wide range of slippery, uneven, rough, or deformable terrains, including wood, glass, ice, soil, grass, and a floating lotus leaf. Owing to fast robot dynamics and the ability to generate large body torques, the robot further demonstrates unparalleled robustness and agility among microrobots. The robot can hop on dynamically inclined surfaces, recover from strong in-air collisions, and perform acrobatic flips in its aerial phase. In addition, diminishing inertial effects enable challenging locomotive tasks that are infeasible for larger-scale robots. Our robot can jump onto a 29-g quadrotor, which shows that impulsive interaction between two heterogeneous aerial robots can enhance mobility. These demonstrations highlight the unique advantages of insect-scale hopping. Our results also have implications on achieving sensing and power autonomy in payload-constrained microrobots that often confront large obstacles. Compared to insect-scale aerial vehicles, our robot retains the capability of overcoming large obstacles while substantially reducing power consumption and increasing payload. The 10 times payload increase opens opportunities for incorporating onboard sensors, electronics, and power sources.

RESULTS

Robot design, hopping dynamics, and control

We proposed two hopping designs (Fig. 1 and fig. S1) for achieving versatile and efficient locomotion in insect-scale robots. Figure 1 (A and B) illustrate two subgram robots that consist of four flapping-wing propulsive modules and a passive monopedal leg. These

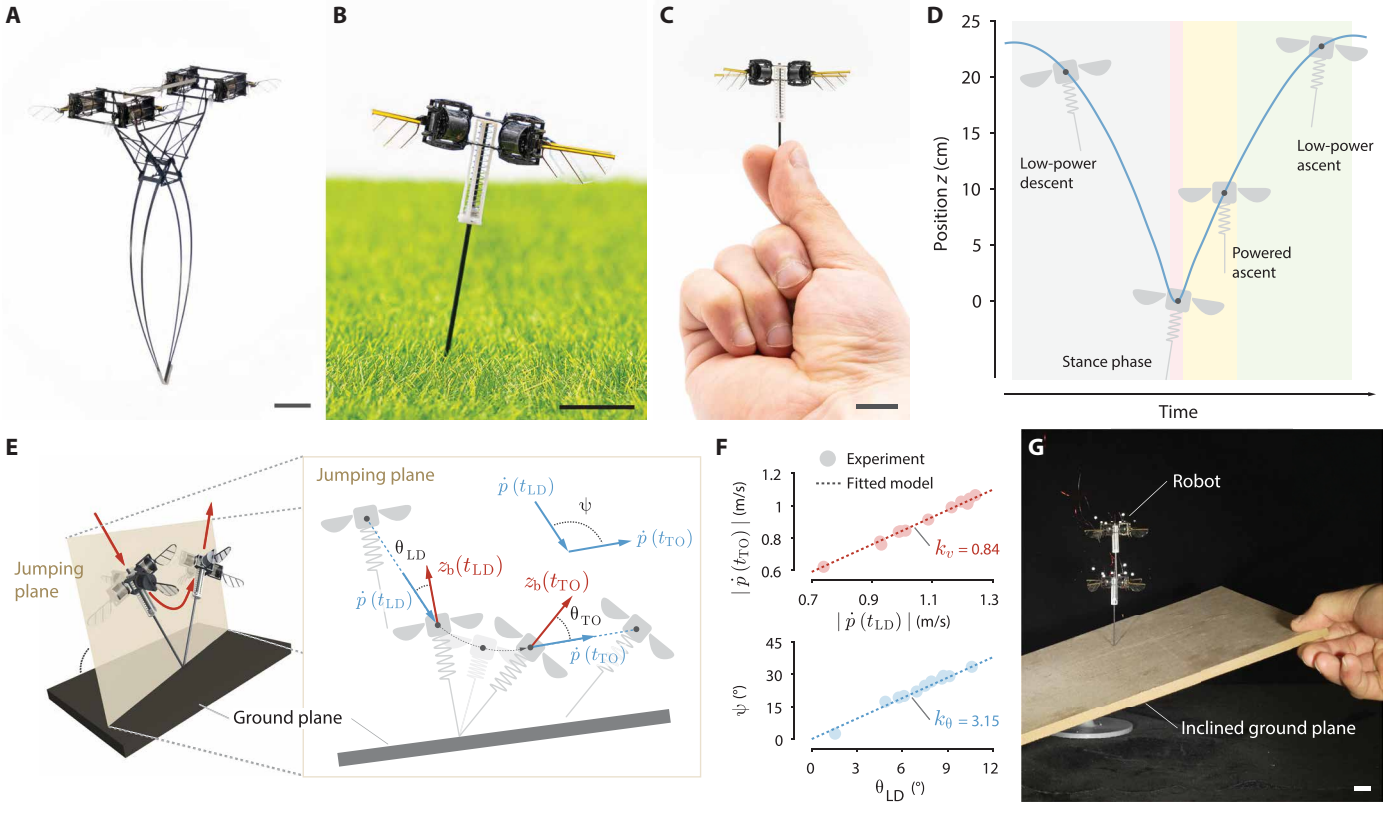


Fig. 1. Hopping design. (**A** and **B**) Images of the eight-wing (A) and four-wing (B) hopping robots. (**C**) Image that illustrates the robot size compared to a human hand. (**D**) Each jumping cycle is split into a short stance phase and a long aerial phase. The aerial phase consists of powered ascent and low-power controlled descent. (**E**) Illustration of a hopping model that relates the landing states $[\mathbf{z_b}(t_{LD}), \dot{\mathbf{p}}(t_{LD})]$ to the takeoff states $[\mathbf{z_b}(t_{TO}), \dot{\mathbf{p}}(t_{TO})]$. In this illustration, θ denotes the angle between robot velocity $\dot{\mathbf{p}}$ and body axis vector $\mathbf{z_b}$, and ψ denotes the rotation between landing and takeoff velocities. (**F**) Empirical relationship between landing and takeoff speeds (top row) and landing and rotation angles (bottom row). (**G**) Composite image of robot hopping on an inclined plate. The scale bars in (A) to (C) and (G) represent 1 cm.

centimeter-scale robots (Fig. 1C) weigh less than 1 g, and their flight modules and leg account for ~80 and 20% of the robot mass, respectively (table S1). The flight modules are driven by power-dense dielectric elastomer actuators (DEAs) that operate in the range of 330 to 400 Hz. In the flight mode, these robots can generate a maximum lift-to-weight ratio of 1.25 and reach a lateral flight speed of 36 cm/s (see the "Robot design and fabrication" section for details on robot performance). Compared to flying, hopping substantially reduces energetic cost while allowing robots to traverse complex terrains that include obstacles, inclines, and surfaces of varying friction. Since insect-scale robots most commonly confront smaller obstacles comparable to their size, it is advantageous for them to efficiently hop over small objects (1.5 to 20 cm) with high frequencies (2 to 8.4 Hz) and precise landing positions and occasionally make high jumps or fly over tall obstacles. Here, we describe mechanical design, modeling, and a feedback controller that enable high-frequency, efficient, and precise hopping.

High-frequency hopping requires fast attitude stabilization and control. Our robots have small moments of inertia, and they can quickly modify forces and torques with four sets of flapping wings. The combination of low inertia and high torque generation allows the two robots (*13*, *25*) to achieve fast body rotation rates of 1800°/s (flapping module in Fig. 1A) and 7200°/s (flapping module in Fig. 1B), respectively. Furthermore, these robots are driven by muscle-like soft actuators that can endure the periodic collisional impact between the leg and the ground. We evaluated two leg designs: a 150-mg, 8-cm-long leg made of compliant carbon fiber laminates (Fig. 1A) and a 220-mg, 5.3-cm-long leg made of a compression spring and 3D-printed casing (Fig. 1B). These lightweight elastic legs showed more than 80% efficiency when the robots were dropped from a 3- to 22-cm height (see the "Jumping leg characterization" section for details on leg performance).

In addition to mechanical designs, hopping requires planning and feedback control (*23*). Hopping locomotion is periodic, and each cycle consists of an aerial phase and stance phase (Fig. 1D). The robot exerts altitude and attitude control in the aerial phase, and it is passive in the 8- to 12-ms stance phase. The aerial phase consists of two parts: short-duration powered ascent and low-power attitude control. During powered ascent that immediately follows the stance phase, the robot compensates for the energy loss from the ground-leg contact by producing lift force. During the rest of the aerial phase, the robot follows a ballistic trajectory and controls landing attitude with ~15% power of free flight. We designed a computationally efficient hopping controller for tracking positional set points ("Hopping dynamics and control" section and fig. S2). At the peak of its trajectory, the robot estimates its landing position $\hat{\mathbf{p}}(t_{\mathrm{LD}})$ on the basis of the instantaneous position $\mathbf{p}(t)$ and velocity $\dot{\mathbf{p}}(t)$. Next, given the desired set point $\mathbf{p}_{\mathrm{D}}$ and the estimated landing position $\hat{\mathbf{p}}(t_{\mathrm{LD}})$, the controller calculates the desired takeoff velocity $\dot{\mathbf{p}}_{\mathrm{D}}(t_{\mathrm{TO}})$ after the stance phase. On the basis of this desired takeoff velocity $\dot{\mathbf{p}}_{\mathrm{D}}(t_{\mathrm{TO}})$, the hopping controller plans for the desired landing attitude $\mathbf{z}_{\mathbf{b}_{\mathrm{D}}}(t_{\mathrm{LD}})$. Figure 1E illustrates the robot's attitude before and after making ground contact. By controlling the offset angle θ_{LD} between the robot's body axis and landing velocity (Fig. 1E), the robot can reach its desired takeoff velocity $\dot{\mathbf{p}}(t_{\mathrm{TO}})$. In our previous work on the 35-g Hopcopter (*20*), the mapping between landing and takeoff states was calculated analytically in real time. This approach has a high computational cost that is unsuitable for high-frequency and low-height hopping. Here, we implemented an empirical mapping where the relationship between the landing $[\mathbf{z}_{\mathbf{b}}(t_{\mathrm{LD}}), \dot{\mathbf{p}}(t_{\mathrm{LD}})]$ and takeoff states $\dot{\mathbf{p}}(t_{\mathrm{TO}})$ was experimentally quantified. Figure 1F shows the linear mapping between input $[|\dot{\mathbf{p}}(t_{\mathrm{LD}})|, \theta_{\mathrm{LD}}]$ and output $[|\dot{\mathbf{p}}(t_{\mathrm{TO}})|, \psi]$ parameters. Since our model does not require the ground plane to be horizontal (Fig. 1E), the robot can hop on inclined surfaces (Fig. 1G) without losing stability or precision (see the "Hopping dynamics and control" section for implementation details).

Precise and efficient hopping demonstrations

We conducted a series of experiments to evaluate robot hopping accuracy, speed, and efficiency (movie S2). On the basis of different hopping tasks, the experiment duration ranges from 3 to 15 s, which is similar to that of flight experiments conducted by subgram robots. First, the four-wing robot was commanded to hop at a set point with fixed altitude. Figure 2A illustrates snapshots of four experiments in which the robot jump heights were set to 4, 7, 10, and 20 cm. For the 4-cm case, the robot hops at ~5 Hz with near-ballistic trajectories. Figure 2 (B to D) shows the robot's tracked *x*, *y*, and *z* positions, respectively. The root-mean-square (RMS) landing position error for the 4-cm hopping experiment is 2.1 cm, and it increases as jump height rises (Fig. 2E). Each hopping experiment in Fig. 2A was repeated to demonstrate robot consistency (figs. S3 to S6). Similar hopping demonstrations were also performed with the eight-wing robot (Supplementary Discussion 1).

Our robot demonstrated the ability to dynamically adjust jump height by changing the duration of powered ascent (movie S2). Figure 2F shows a hopping experiment in which the desired altitude increased from 4 to 10 cm in increments of 3 cm. The tracked robot trajectory (Fig. 2, G and H) was approximately ballistic where the RMS altitude and position errors were 0.7 and 4.2 cm, respectively. Since the absolute landing error positively correlates with jump height (Fig. 2E), our low-height and high-frequency hopping design showed higher absolute landing accuracy compared to existing hopping systems (*19*, *20*). The robot's relative landing accuracy, as measured by the ratio between the landing position error and the jump height, was in the range of 0.3 to 0.5 and comparable to that of prior systems (*19*, *20*). Our robot could accurately track set points (Fig. 2, I to K) and overcome obstacles (Fig. 2, L to N). When the robot hopped from one set point to another (Fig. 2, I to K, and movie S3), it reached an average lateral speed of 32 cm/s—comparable to its flight speed of 36 cm/s (*25*). With a hopping height of 7 cm, the RMS position error was 2.9 cm (Fig. 2K). In addition, the robot could jump on and off a 5.5-cm-high obstacle (Fig. 2, L to N, and movie S3). As the obstacle had a planar dimension of 15.5 cm by 10.5 cm, a high landing accuracy was required for the robot to alternate between the ground and the elevated obstacle surface. The tracked robot altitude and body axis angle are shown in Fig. 2 (M and N), which highlights rapid adjustments of hopping height and body axis. The experiments in Fig. 2 (F, I, and L) were repeated (figs. S7 to S9) to demonstrate hardware and controller consistency. Although variable height hopping enables obstacle traversal as demonstrated in Fig. 2L, it is limited to overcoming stair-like flat obstacles. Since landing error increases as jump height increases, the robot would face difficulty landing on a tall and thin object. To improve landing accuracy when performing high jumps, the robot needs to exert position control during the descent phase, which would reduce jumping efficiency.

Compared to flight, hopping locomotion substantially improved robot payload and reduced power consumption. While our robot

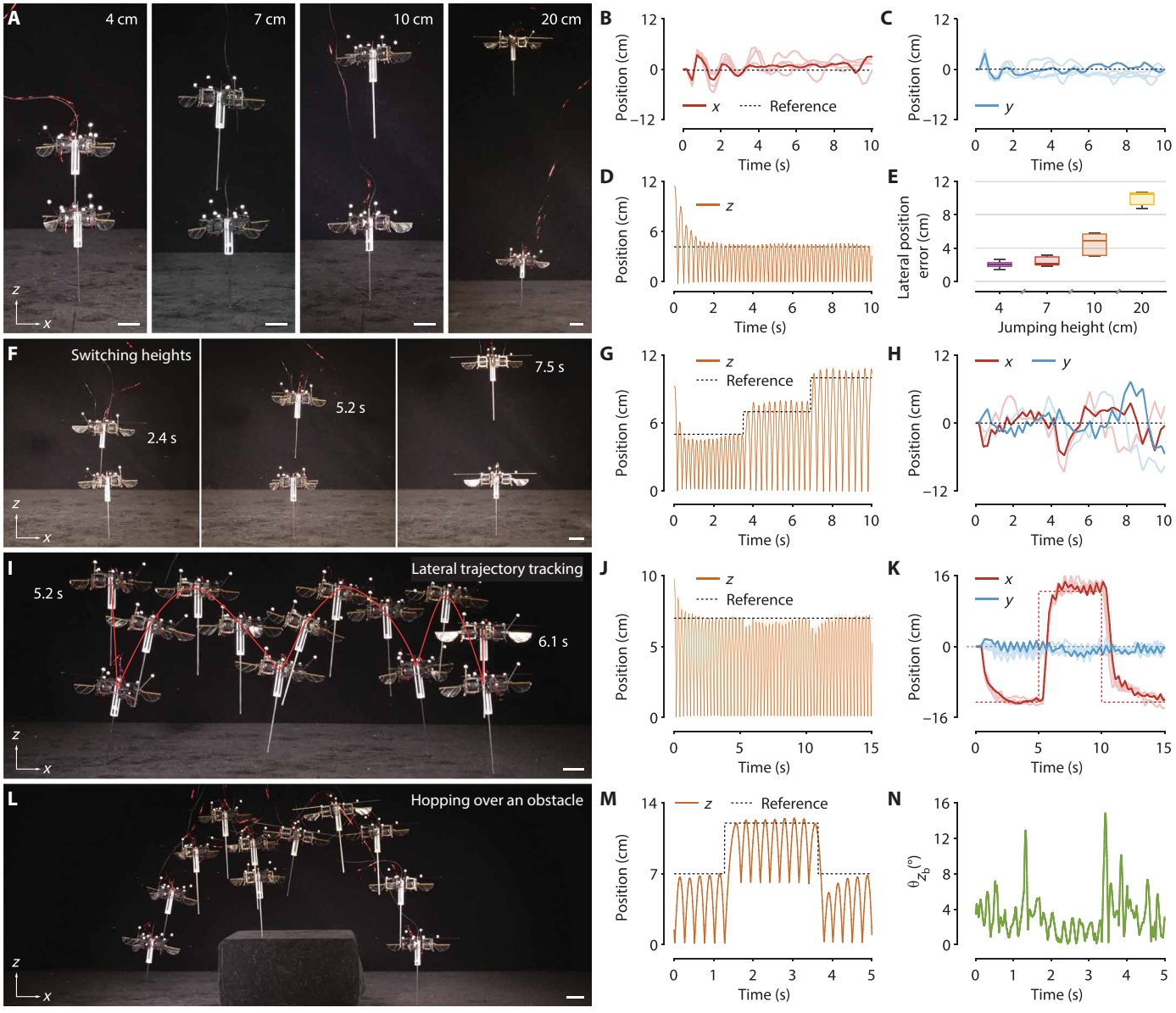


Fig. 2. Robot hopping demonstrations. (**A**) Composite images of four hopping experiments with heights of 4, 7, 10, and 20 cm. (**B** to **D**) Tracked *x*, *y*, and *z* positions for the 4-cm hopping experiments. (**E**) The mean landing position error increases as a function of hopping height. (**F**) Composite image sequence of an experiment where the jumping height changes from 4 to 10 cm in increments of 3 cm. (**G** and **H**) Tracked altitude (G) and lateral (H) positions corresponding to the experiment in (F). (**I**) Composite image of a set point switching experiment. (**J** and **K**) Tracked altitude (J) and lateral (K) positions corresponding to the experiment in (I). (**L**) Composite image of an obstacle traversal demonstration. (**M** and **N**) Tracked altitude (M) and lateral (N) positions corresponding to the experiment in (L). The scale bars in (A), (F), (I), and (L) represent 1 cm. The darker-colored curves in (B), (C), (H), and (K) correspond to the experiments shown in (A), (F), and (I), and the lighter-color curves represent repeating trials.

could carry at most 200 mg in flight (*25*), its payload capacity easily reached 2 g during hopping (Fig. 3A, fig. S10, and movie S3). This substantial increase in payload is partially contributed by the reduction of force generation required by hopping. As an example, Fig. 3B shows the commanded thrust force normalized by robot weight. When the robot hopped without payload (yellow curve in Fig. 3B), the controller commanded 64% thrust force (F/mg) during the powered ascent phase that accounted for 21% of the hopping cycle. During the entire aerial phase, the robot also generated attitude control torques by differentially changing the forces among its four modules (Fig. 3C). The net lift forces, which included contribution from thrust and flight torque production, showed a substantial reduction compared to hovering flight (Fig. 3D). Without carrying payload, the robot required less than 20% of nominal lift forces to hop between 4 and 10 cm. With a 2-g payload that is more than 200% of the robot's weight, the robot only required 33% nominal lift force. We estimate that the robot's maximum payload capacity would exceed 7 g. Figure 3D also shows that mean lift force increased as hop height increased, which suggests that high-frequency and low-height hopping consumes less power than making high jumps.

We further estimated robot input power consumption for the same set of hopping experiments. Figure 3E shows a force-to-power

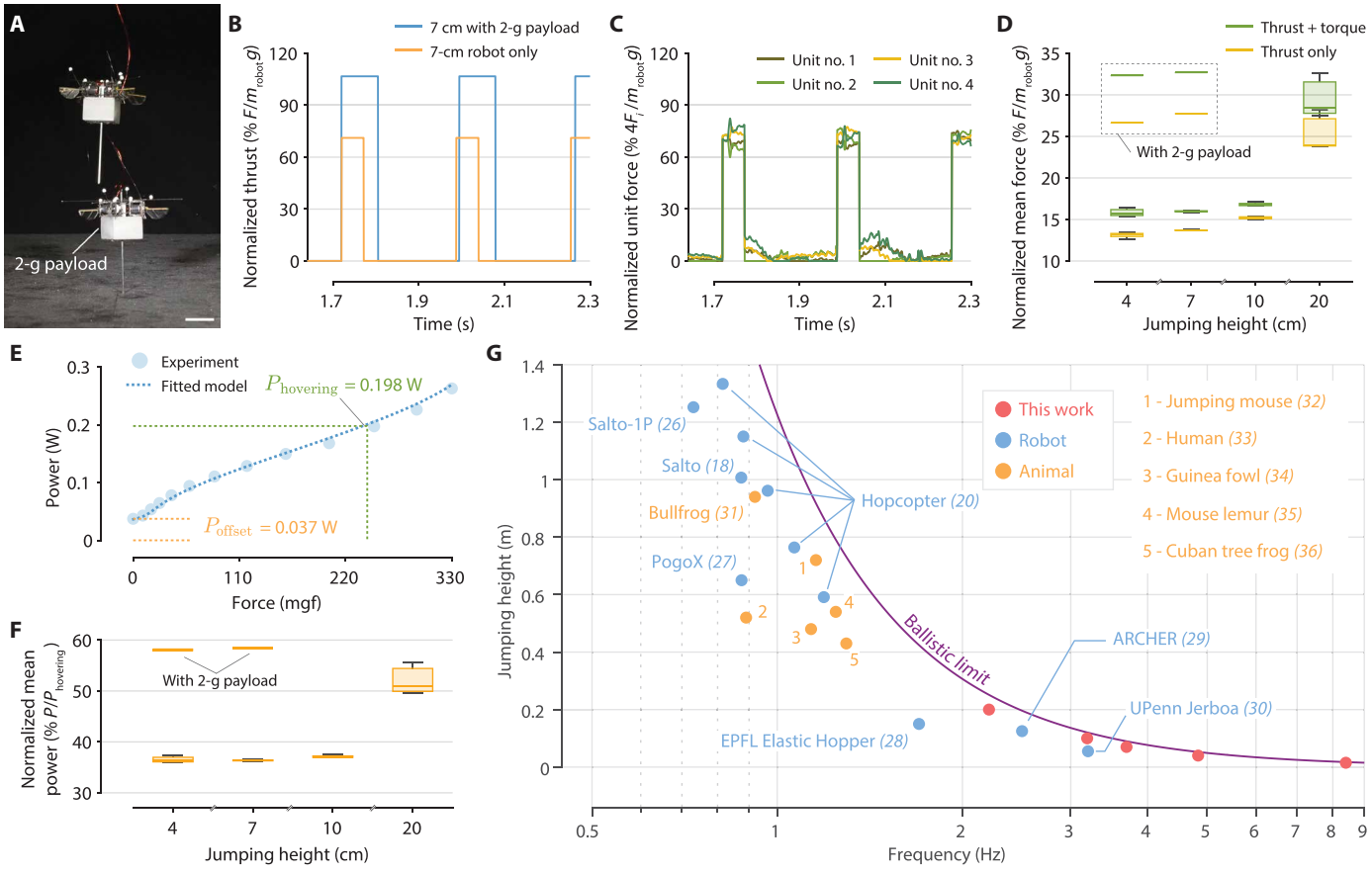


Fig. 3. Efficiency and payload capacity of low-height, high-frequency hopping. (**A**) Composite image of 7-cm robot hopping while it carried 2-g payload. (**B**) Commanded thrust force normalized by robot weight. (**C**) Commanded lift forces for each robot module that correspond to the 7-cm hopping (no payload) case in (B). (**D**) Normalized net lift forces for various hopping height and payload conditions. The net lift force includes contributions from thrust and control torques. (**E**) Force to input power mapping for a robot module. (**F**) Normalized power consumption for various hopping height and payload conditions. (**G**) Comparison of jump frequency and height among existing hopping robots and animals (*18*, *20*, *26*–*36*). The purple curve represents the ballistic limit as an upper bound. The ARCHER robot's hopping height was estimated on the basis of videos and graphs from a prior work (*29*).

mapping based on static flapping experiments. The mapping is nonlinear under low thrust conditions, and there exists a power offset under the zero-lift-force condition. This offset accounts for 19% of power consumption under the hovering condition, and it is contributed by nonlinear wing stroke and pitch coupling. On the basis of this mapping, we compared hopping power consumption with respect to hovering flight (Fig. 3F). While the normalized power (P / P_{hovering}) was ~20% higher than the normalized lift (F_L / mg) (Fig. 3D), this result still represented a substantial power saving in the 45 to 64% range compared to flight. This trend was supported by our model (Supplementary Discussion 2) predicting that power consumption is nominally proportional to the square root of hopping height h. However, the energetic benefit diminishes at very low hop heights because of a combination of attitude control effort and the constant power offset. While the power offset could be reduced by completely switching off a robot module when commanding zero lift force, the attitude control effort for each hop remains constant.

We further compared flight and hopping energetic efficiency on the basis of these results. Hovering efficiency is measured by the lift-to-power ratio. Under this metric, our hopping design showed a 200% increase in effective lift and more than 40% power reduction, equivalent to 3.3 times the lift-to-power ratio of hovering. We also compared the robot COT during lateral flight and set point switching demonstrations. Under a flight speed of 36 cm/s and a hopping speed of 32 cm/s, the robot flight and hopping COT were 2.57×10^3 and 1.27×10^3, respectively. This result highlights the efficiency improvement of hopping locomotion. However, varying hop height does not reduce COT if the robot maintains a constant landing angle θ_{LD} for avoiding foot slippage. Under a fixed landing angle, the robot's lateral speed is proportional to the square root of jump height. The power saving that resulted from lower jump height would be counteracted by a reduction of lateral speed. We derived a model (Supplementary Discussion 2) predicting that COT is independent of hopping height.

Our robot demonstrated the highest hopping frequency compared to robots and animals (*26*–*36*) shown in Fig. 3G. We only accounted for intrinsically unstable hopping systems that require feedback control, excluding bristlebot-like vibration-based ambulatory robots (*37*). The relationship between jump height and frequency is bounded by the ballistic limit $f = \sqrt{g / 8h}$ (Supplementary Discussion 2), where f is the frequency and g is the gravity constant. Owing to smaller inertia and higher control bandwidth, our robot

could perform hopping with 1.6 to 10.5 times higher frequency than these existing platforms. The primary benefit of low-height and high-frequency hopping relates to the reduction of absolute landing error. With higher absolute landing precision, smaller robot-to-ground impact, and faster attitude control, our robot is adaptive to hopping on a wide range of challenging terrains.

Hopping adaptability and agility

Owing to their small size, terrestrial microrobots face challenges when they traverse terrains of varying roughness, surface textures, and incline angles. For instance, a legged microrobot may slip on an icy surface, and its legs may be entangled when it crawls on grass or soil. In contrast, our hopping microrobot can effectively traverse these surfaces. To satisfy the friction cone condition ($F_f \leq \mu F_N$) on a slippery surface, the robot can reduce its lateral landing velocity v_l under the relationship $\tan(v_l / v_z) < \mu$. In addition, maintaining a small landing offset angle θ_{LD} can reduce robot rotation during the stance phase (Fig. 1E), which reduces the probability of leg entanglement for hopping on textured terrains such as grass and soil. If the surface is soft and dampening, the robot can compensate for the energy loss in the stance phase by extending powered ascent duration.

To evaluate locomotion adaptability, we demonstrated hopping on a variety of surfaces including wood, ice, glass, soil, and grass (movie S4). Figure 4A shows snapshots of these experiments, and figs. S11 to S15 show the tracked robot position and attitude. These experiments were repeated five times to highlight robot and controller reliability. Our hopping controller adaptively adjusted the net commanded lift force on the basis of stance phase energy recovery from different surfaces. Figure 4 (B and C) shows that 83 and 61% more thrust and torque were needed for the robot to jump on grass compared to glass, respectively. Since a single controller could handle all these terrains, the robot easily transitioned from one surface to another. Figure 4D shows a set point switching demonstration from grass to wood. The tracked robot position and altitude (Fig. 4, E and F) show similar speed and accuracy compared to hopping on one surface (Fig. 2I). This experiment was repeated five times (fig. S16).

In addition, our robot is robust against dynamic disturbances. In a hopping experiment (Fig. 5A and movie S5), we dynamically tilted a wooden plate to vary the surface incline angle. The robot's landing altitude changed as a function of plate's inclination and lateral landing position. Without knowledge of surface rotation, the robot maintained a 5.3-cm hopping height measured with respect to the fixed lab coordinates (Fig. 5, B and C). Since the landing altitude

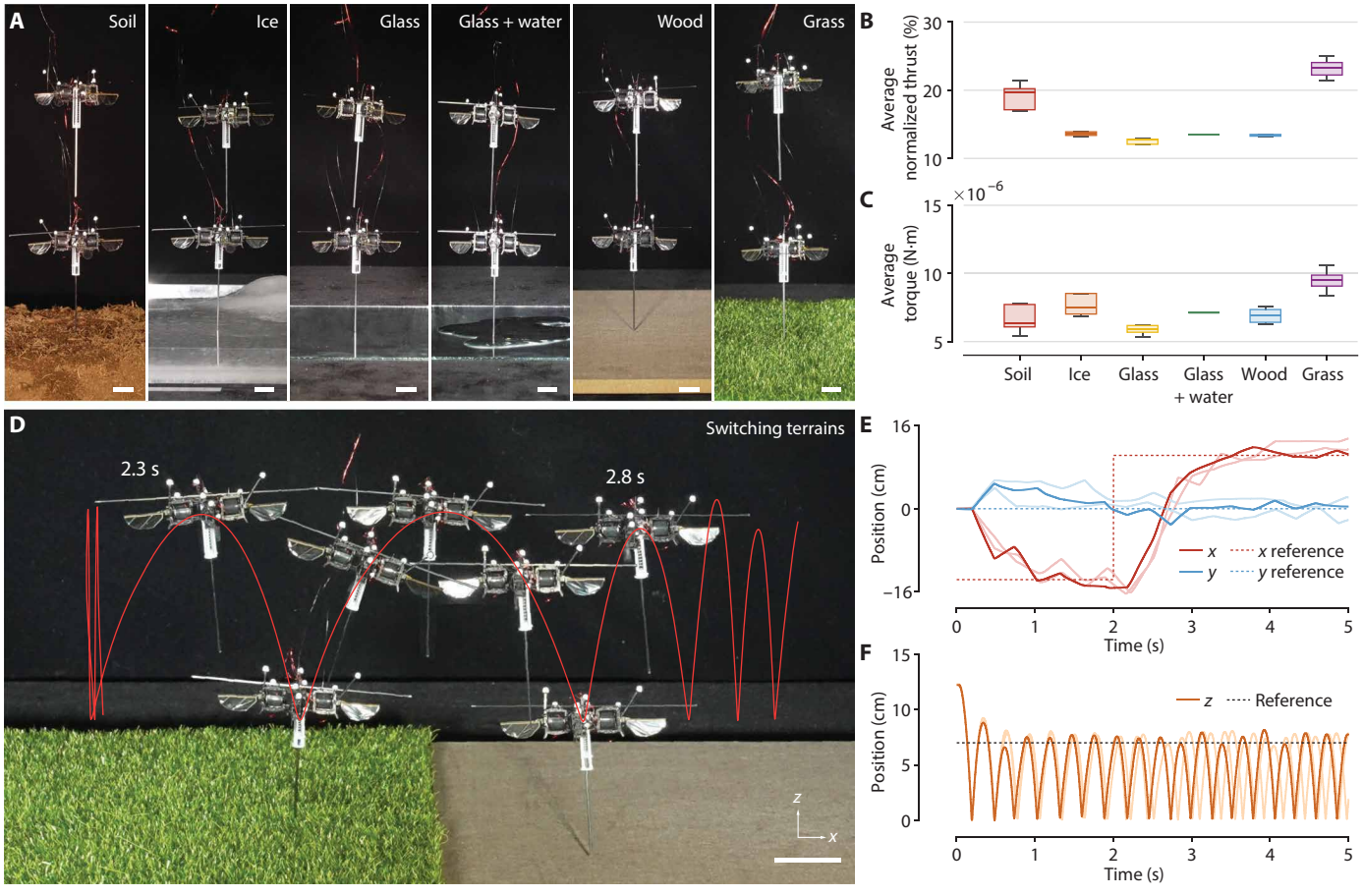


Fig. 4. Hopping demonstrations on different terrains. (**A**) Robot hopping demonstrations on a variety of surfaces including soil, ice, glass, glass with water, wood, and grass. (**B** and **C**) Average thrust and control torque corresponding to the experiments in (A). (**D**) Robot set point switching demonstration from grass to wood. (**E** and **F**) Tracked lateral position (*x*, *y*) and altitude (*z*) corresponding to the experiment in (C). The scale bars in (A) and (D) correspond to 1 cm. The lighter-colored curves in (E) and (F) correspond to the repeated experiments.

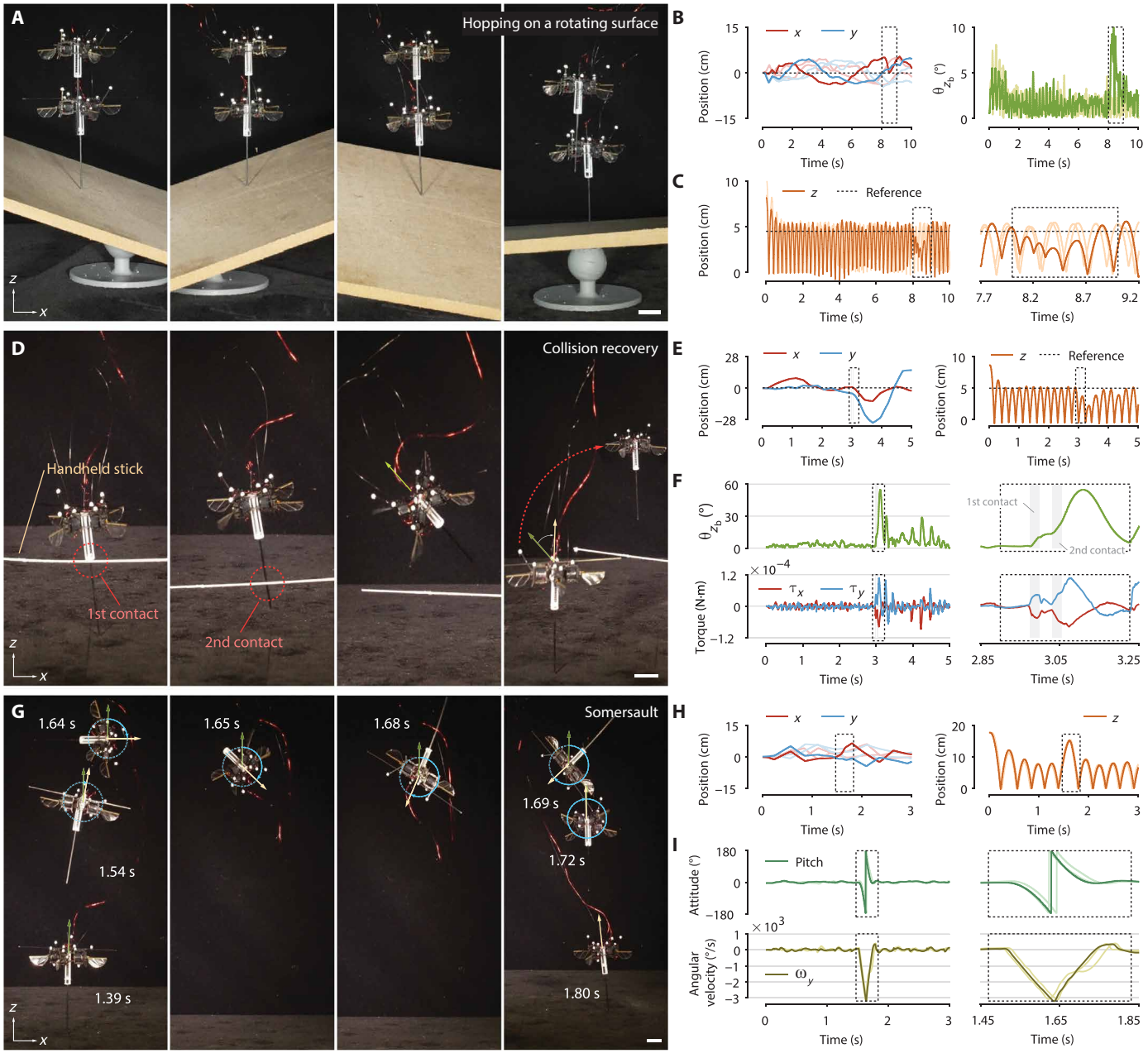


Fig. 5. Hopping robustness and agility. (**A**) Composite image sequence of robot hopping on a dynamically rotating plate. (**B** and **C**) Tracked position, altitude, and body axis angle corresponding to the experiment in (A). (**D**) Composite image sequence of hopping collision recovery. (**E** and **F**) Tracked position, altitude, and body axis angle corresponding to the experiment in (D). (**G**) Composite image sequence of robot somersault between consecutive jumps. (**H** and **I**) Tracked position, altitude, and body axis angle corresponding to the experiment in (G). In (B), (C), (E), (F), (H), and (I), the darker-colored curves represent the experiments shown in (A), (D), and (G), respectively. The lighter-colored curves represent repeating experiments. The scale bars in (A), (D), and (G) represent 1 cm. The experiments in (A), (D), and (G) were repeated and are shown in figs. S17 to S19.

changed but the desired hopping height remained unchanged, the relative vertical hopping distance varied between each hop. Consequently, the controller modified forces and torques in the aerial phase, which led to changes of hopping frequency. In this experiment, the robot hopping frequency and relative hop height varied between 4.2 and 8.4 Hz and between 1.5 and 6.4 cm, respectively. The boxed region in Fig. 5C highlights quick changes of body axis angle and hopping height because of surface rotation.

Owing to its small inertia and the ability to produce large body torques, our robot can recover from strong collisions during hopping (see Supplementary Discussion 3 for scaling analysis on fast attitude control). Figure 5D and movie S5 show an experiment where the robot was hit with a stick. At a height of 3 cm, two consecutive collisions caused a sudden increase in the robot's translational and angular speeds (Fig. 5E). The hopping controller prioritized attitude stabilization by commanding a restoring torque of 6.1×10^{-5} N·m.

As a result, the attitude deviation reduced from 54.8° to 5.2° within 113 ms, and then the robot gradually recovered to its position set point within six hopping periods (Fig. 5, E and F).

In addition to rejecting dynamic disturbances, our robot could demonstrate acrobatic maneuvers such as hopping-assisted body flips (Fig. 5, G to I, and movie S5). Flight acrobatics are challenging for subgram robots because they require high torques, high lift forces, and high-rate closed-loop control. In the past, while a 0.7-g aerial robot could demonstrate a body flip, it could not recover altitude stability before hitting the ground (*13*). The robot-ground collision dissipated energy and required high lift forces for altitude recovery. In contrast, hopping allows a robot to efficiently perform somersaults without requiring large lift forces. Figure 5G shows an image sequence of the robot completing a somersault between two hopping cycles. In this process, the robot generated a maximum normalized lift force (F_L / mg) of 0.93—merely 45% higher than nominal hopping in Fig. 2A. It performed the somersault in 0.13 s at a 15.1-cm hopping height and repeated this maneuver five times (fig. S19). This challenging demonstration highlights insect-like efficiency, agility, and robustness in a subgram hopping robot.

Furthermore, owing to its small inertia, our robot could perform unique tasks that are challenging for larger robots. While most aerial robots avoid in-air collisions, our microrobot could hop on top of a palm-sized drone. In the hopping experiment illustrated in Fig. 6A and movie S6, our 0.97-g robot was required to move across a 19-cm gap. The robot was unable to fly over the gap because its normalized lift force (F_L / mg) was virtually limited to 0.93. In this collaborative demonstration, a 29-g quadrotor (Bitcraze Crazyflie 2.1) carried a flat surface on its top and flew upward to bridge the gap. Next, our robot traversed the gap through hopping on top of the quadrotor. As the robot landed on and bounced off the large quadrotor, the collisional impact caused attitude disturbances in both platforms (Fig. 6B). The hopping robot had faster dynamics and could quickly stabilize its attitude, whereas the quadrotor experienced smaller attitude change thanks to its substantially larger inertia. The hopping robot position, altitude, and orientation are shown in Fig. 6C. The shaded region in Fig. 6C highlights the duration of robot impact on the hovering quadrotor, and it shows attitude recovery within 200 ms. This demonstration showcases that the strategic use of impulsive interactions between differently sized machines could enhance locomotion capability. Such a technique would likely be infeasible if both platforms had comparable weights. The robot's small mass also enabled interactions with delicate objects such as floating lotus leaves (movie S6 and Fig. 6, D and E). While larger hopping robots may damage the lotus leaf and fall into water, our robot repeatedly hopped on the leaf while inducing small ripples on the water surface. Figure 6E shows the corresponding robot position and altitude, and the experiment was repeated five times (fig. S20). These demonstrations illustrate that low inertia, fast control, and collision robustness enable unique capabilities in small hoppers. It opens opportunities for studying collision-friendly behaviors within a heterogeneous robot swarm, as well as extending robot locomotion capabilities to a wide range of delicate and challenging terrains.

DISCUSSION

In this work, we presented a subgram hopping robot capable of efficiently traversing complex terrains. In contrast to most hopping designs, the robot was equipped with four sets of flapping wings for generating lift forces and control torques in the aerial phase. The robot demonstrated low-height and high-frequency hopping in the ranges of 1.5 to 20 cm and 2 to 8.4 Hz, with a 10-fold increase in maximum payload capacity and 51% reduction in COT. Under feedback control, the robot could track set points, overcome tall obstacles, and hop on a wide range of surfaces including wood, glass, ice, soil, and grass. The robot exhibited outstanding robustness and agility owing to its small inertia and large torque generation capability. It could hop on a dynamically inclined surface, recover from strong in-air collisions, and perform somersaults between consecutive jumps. These hopping demonstrations have never been achieved in subgram robots. In addition, our robot could perform tasks unique to lightweight insect-scale systems such as jumping on a larger quadrotor or hopping on a floating lotus leaf.

These locomotion demonstrations were enabled by a combination of intelligent mechanical and algorithmic designs. From the hardware perspective, high-frequency hopping poses unique challenges because microscale systems have fast dynamics and require even faster force and torque generation. It is impractical to exert force and torque control during the short (<20 ms) stance phase when the leg contacts the ground. We addressed this challenge by incorporating a passive leg with actively controlled wings. In contrast to most existing designs (*38*–*41*), our robot exerts attitude control in the longer (>200 ms) aerial phase and uses a passive leg to efficiently (>80%) recover energy during the stance phase. Our power-dense DEAs enabled high-frequency (330 to 400 Hz) flapping-wing propulsion that can quickly produce control forces and torques. In particular, the quick transition between powered ascent and low-power control (Fig. 3C) requires rapid switching of DEA operating conditions that induce large internal stress. Rigid motors and piezoelectric ceramics cannot operate under such high stress and high frequency condition owing to limited bandwidth and robustness. The use of soft actuators and compliant mechanisms further ensures robot-level collision resilience—a rare property among rigid-driven aerial robots. To demonstrate hopping, these intelligent hardware designs also require planning, estimation, and control. We designed a computationally efficient controller that can estimate landing position, plan and control landing attitude, and adaptively adjust jumping altitude. The combination of these hardware and controller designs enables efficient, robust, and agile hopping.

Our work also highlights the importance of achieving soft robotic agility, efficiency, and robustness at the microscale. While most soft robotic studies focus on robustness, safety, and adaptability, soft systems often have limited speed, precision, and controllability. Consequently, it remains difficult for soft robots to traverse challenging terrains, overcome large obstacles, and perform insect-like agile maneuvers. In contrast, rigid driven microrobots (*11*, *42*, *43*) are precise, fast, and controllable, and they can fly (*11*), jump (*42*), and run (*43*) with insect-like performance. However, unlike insects that can robustly interact with environments, rigid driven robots may experience damage when they bump into a flower or fall off from a shelf. To find insect-like applications, small robots must be simultaneously agile and robust. Our hopping robot fits these criteria because it can traverse challenging terrains while surviving large impulses caused by periodic leg-ground contact. The combination of power-dense soft actuators, compliant mechanisms, and rigid appendages has enabled agile and robust locomotion that has not been demonstrated in prior robots.

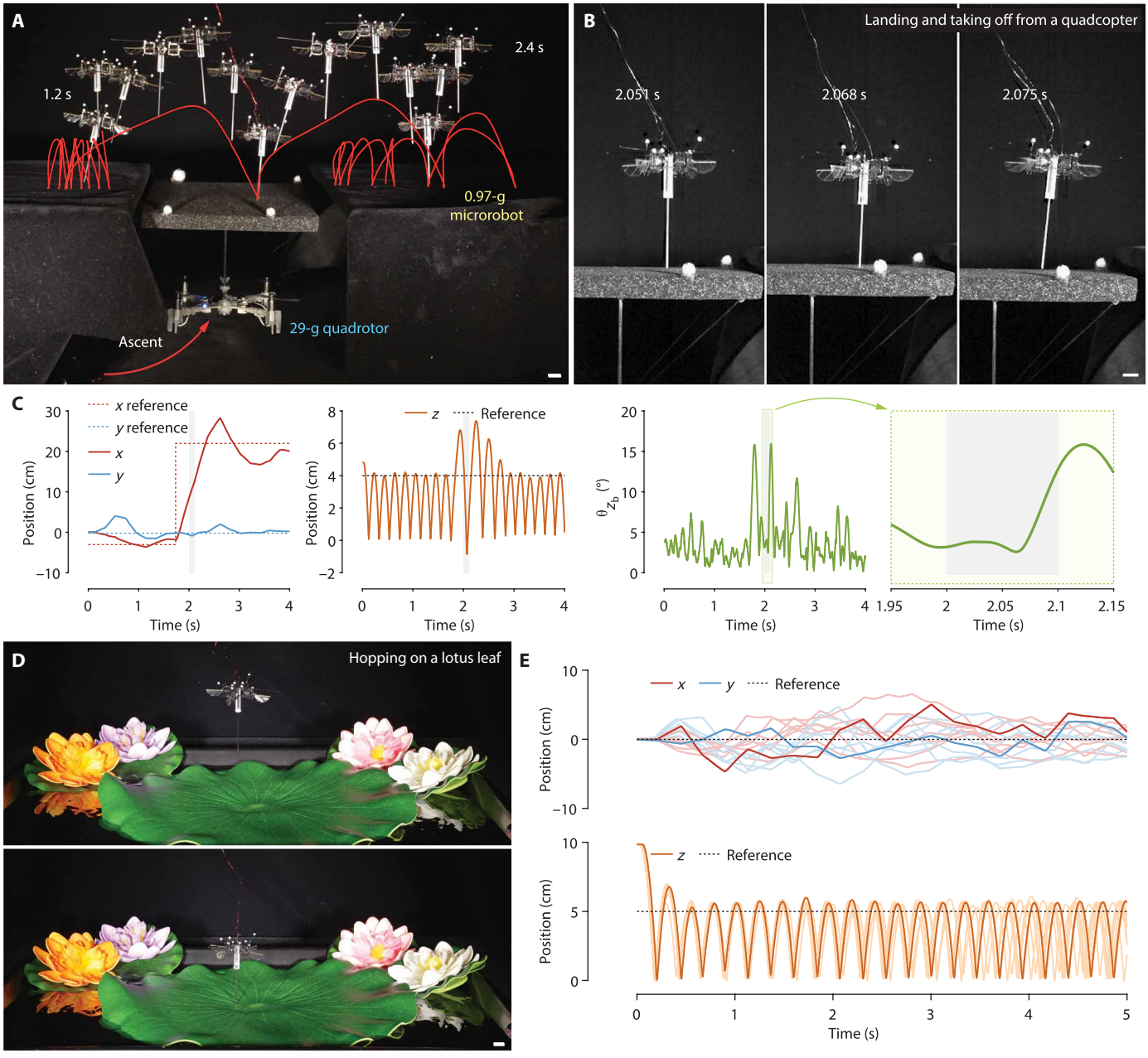


Fig. 6. Hopping demonstrations on airborne and compliant objects. (**A**) Composite image of the robot moving across a 19-cm gap by hopping on a 29-g quadrotor. (**B**) Image sequence of the robot before, during, and after impacting the hovering quadrotor. (**C**) Tracked robot position, altitude, and body axis angle corresponding to the experiment in (A). (**D**) Image sequence of the robot hopping on a floating lotus leaf. (**E**) Tracked robot position, altitude, and body axis angle corresponding to the experiment in (D). The lighter-colored curves show data from repeated experiments. In (A), (B), and (D), the scale bars represent 1 cm.

In addition to demonstrating capabilities that are absent in existing subgram robots, the hopping design also shows a substantial improvement of energetic efficiency and payload—an important result that will benefit future works on power autonomy. During flight, the robot consumes 0.8-W power with under 200-mg payload, which is inadequate for incorporating sensors, power electronics, and batteries. In contrast, insect-scale terrestrial robots have demonstrated various degrees of autonomy (*44*) owing to larger payload and lower COT. Since our hopping design reduces power requirement and increases payload, it may leverage existing autonomy solutions in similar-sized terrestrial systems (*45*, *46*). Under hopping, our robot consumes 0.47 W while carrying 2-g payload. Given this payload capacity, sensors, boost electronics, and batteries could account for 0.3, 0.7, and 1 g, respectively. On the basis of the energy density of lithium battery (200 W·h/kg) and the transduction efficiency (10 to 20%) of our prior boost electronics (*47*), we expect that the robot would hop for 2 to 5 min under this energetic and payload budget. This analysis opens opportunities for developing autonomous microrobots that can traverse challenging terrains.

METHODS

Robot design and fabrication

The jumping robot designs consist of flapping-wing aerial robots and passive telescopic legs. The four-wing robot (fig. S1A) has an elastic leg made of steel spring, a 3D-printed case, and a carbon fiber rod (fig. S1B). The leg becomes compressed during the stance phase and returns to a relaxed phase during the aerial phase (fig. S1C). Figure S1D shows that the leg is compressed by 5 to 8 mm during the stance phase. The dimension and mass of this leg design are 0.5 cm by 0.5 cm by 5.3 cm and 220 mg, respectively. In comparison, the eight-wing robot (*12*) (fig. S1E) has larger dimensions and smaller net payload. We designed a lightweight elastic leg that consists of tapered carbon fiber laminates bent to form a passive spring (fig. S1, F to H). Figure S1I shows a robot hopping experiment where the leg is also compressed by 5 to 7 mm. We also tested a similar leg design (fig. S1, J to N) where the spring was shorter and stiffer. The fitted performance parameters for these two carbon fiber legs are shown in fig. S1 (O and P). In comparison to the steel spring leg, the carbon fiber legs have larger dimensions and lighter mass. Their dimensions and mass are 3.3 cm by 1.0 cm by 8.0 cm and 150 mg, respectively. However, these legs exhibit more off-axis motion during the stance phase, leading to larger variation of takeoff speed and angle. Consequently, the four-wing robot demonstrates higher landing accuracy compared to the eight-wing robot. Table S1 documents the leg design parameters.

The robots are fabricated using existing methods described in prior work (*48*). The two telescopic legs are fabricated on the basis of 3D and laser micromachining printing methods. The steel leg design consists of two 3D-printed components serving as the casing and guide rail of the spring (fig. S1, B and C). We used a resin printer (FormLabs Form 3) with the Rigid 10K material. A steel spring (Uxcell 304 stainless steel 0.2 mm) and a 1-mm-diameter carbon fiber rod are inserted into the 3D-printed parts. The carbon fiber rod is rigidly affixed to the lower 3D-printed component. The telescopic leg is constrained to retract along the axial direction with an effective stiffness of 72 N/m. This design is compact and fits within the center of the four-wing robot.

The carbon fiber leg (fig. S1) consists of seven to nine pieces of laser-micromachined components (fig. S1, F and K). All laser-cut components are made of 160-μm-thick carbon fiber, and they are hand assembled under a confocal microscope. Two planar carbon fiber components are curved to form a preloaded spring, which has an effective stiffness of 122 N/m. This structure is mounted to the base of the eight-wing robot with glue (Loctite 416).

Jumping leg characterization

To characterize the leg efficiency and the stance phase dynamics, we conducted drop tests to record robot landing and takeoff states. For the prototype with a compression spring leg, the robot was dropped from a height of 2.8 to 7.9 cm for 12 times at various orientations slightly deviated from the vertical. The landing and takeoff states (attitude $\mathbf{z_b}$ and velocity $\dot{\mathbf{p}}$) were measured using the motion capture system at a sampling rate of 400 Hz. We obtained the landing speeds and takeoff speeds in the ranges of 0.74 to 1.24 and 0.62 to 1.06 m/s, respectively, with an average stance duration of 10 ± 2 ms. The robot landed with its attitude and velocity direction deviated from the vertical by 3.8° to 9.7° and 0.8° to 10.7° on average.

Figure 1F shows the relationship between the robot's landing speed $\|\dot{\mathbf{p}}(t_{\mathrm{LD}})\|$ and takeoff speed $\|\dot{\mathbf{p}}(t_{\mathrm{TO}})\|$. The results suggest a linear relationship, with the best fit coefficient $k_v = 0.84$. Figure 1F also shows the relationship between landing offset angle θ_{LD} and angular displacement ψ during the stance phase. θ_{LD} is defined as the angle between the robot body axis $\mathbf{z_b}(t_{\mathrm{LD}})$ and its velocity $\dot{\mathbf{p}}(t_{\mathrm{LD}})$ at the moment of landing (Fig. 1E), and ψ is the rotational angle between $\mathbf{z_b}(t_{\mathrm{LD}})$ and its velocity $\dot{\mathbf{p}}(t_{\mathrm{TO}})$ at the moment of takeoff. The best fit coefficient k_θ between the variables θ_{LD} and ψ is 3.15. On the basis of these two empirical relationships, the takeoff velocity $\dot{\mathbf{p}}(t_{\mathrm{TO}})$ can be predicted with landing velocity $\dot{\mathbf{p}}(t_{\mathrm{LD}})$ and landing angle θ_{LD}.

The same measurements were taken for the eight-wing robot with a compliant carbon fiber leg (see fig. S1). In these experiments, the robot was dropped from a height of 11.5 to 21.9 cm for 32 times, with best fit k_v and k_θ of 0.91 and 2.35, respectively.

Hopping dynamics and control

The hopping cycle of our robot is dominated by the aerial phase, accounting for more than 90% of the total time. During this phase, the robot's dynamics mirror those of insect-like flapping-wing robots (*12*, *13*), where the system can be modeled as a rigid body under gravity. The dynamics are primarily influenced by thrust T and torque τ, with air drag considered negligible at this scale.

Unlike typical aerial robots, our hopper operates under a near-free-fall condition for most of the aerial phase, producing minimal thrust and torque. The hopping trajectory is controlled through two key mechanisms, (i) height regulation via a brief powered ascent and (ii) attitude control at landing, which determines the takeoff direction and velocity for the subsequent aerial phase as well as the subsequent landing location.

Height regulation

Height control is achieved by commanding the duration of powered ascent. Immediately after takeoff from the stance phase, the robot generates thrust by flapping its wings. To reach a desired height h_{D}, we command the robot to assume an upright attitude ($\mathbf{z_b} \rightarrow \mathbf{e_3}$) and generate a total lift of δmg (where $\delta < 1$). Concurrently, we monitor the total mechanical energy $mg\mathbf{e_3}^T\mathbf{p} + \frac{1}{2}m\dot{\mathbf{p}}^T\dot{\mathbf{p}}$. Once this energy reaches the target potential energy mgh_{D}, the controller transitions to a low-power mode where it tracks the desired landing attitude.

Lateral set point tracking

We designed a hopping controller for the robot to efficiently track lateral position set points. The robot follows an approximately ballistic trajectory in every hopping cycle, and the estimation, planning, and control process consists of six sequential steps. In describing the controller design, we use the "hat" notation to denote estimated quantities.

First, after the robot completes powered ascent, it estimates the landing position $\hat{\mathbf{p}}_{\mathrm{LD}}$ and velocity $\hat{\dot{\mathbf{p}}}_{\mathrm{LD}}$, assuming that it follows a ballistic trajectory (fig. S2A). Given the instantaneous position $\mathbf{p}$ and speed $\dot{\mathbf{p}}$, the landing position $\hat{\mathbf{p}}_{\mathrm{LD}}$ and velocity $\hat{\dot{\mathbf{p}}}_{\mathrm{LD}}$ are estimated as

$$\hat{\mathbf{p}}_{\mathrm{LD}} = \mathbf{p} + \dot{\mathbf{p}}\hat{t} - \frac{1}{2}g\hat{t}^2\mathbf{e_3} \tag{1}$$

$$\hat{\dot{\mathbf{p}}}_{\mathrm{LD}} = \dot{\mathbf{p}} - g\hat{t}\mathbf{e_3} \tag{2}$$

where $\hat{t}$ is the estimated time to landing, and $\mathbf{e_3}$ is the z-direction unit vector. $\hat{t}$ can be predicted with the following equation

$$\mathbf{e_3}^T\hat{\mathbf{p}}_{\mathrm{LD}} = \mathbf{e_3}^{\mathrm{T}}(\mathbf{p} + \dot{\mathbf{p}}\hat{t} - \frac{1}{2}g\hat{t}^2\mathbf{e_3}) = 0 \tag{3}$$

In the second step (fig. S2B), the algorithm plans for the next landing position $\mathbf{p}_{\mathrm{PLAN,LD}}$ on the basis of the final desired landing location $\mathbf{p}_{\mathrm{D}}$ and immediate landing position $\hat{\mathbf{p}}_{\mathrm{LD}}$. It attempts to incrementally reduce the landing position error $\mathbf{p}_{\mathrm{D}} - \hat{\mathbf{p}}_{\mathrm{LD}}$ under the following equation

$$\mathbf{p}_{\mathrm{PLAN,LD}} = \hat{\mathbf{p}}_{\mathrm{LD}} + \gamma(\mathbf{p}_{\mathrm{D}} - \hat{\mathbf{p}}_{\mathrm{LD}}) \quad (4)$$

where γ is an incremental gain that determines the controller convergence rate. The relationship between $\mathbf{p}_{\mathrm{PLAN,LD}}$, $\hat{\mathbf{p}}_{\mathrm{LD}}$, and $\mathbf{p}_{\mathrm{D}}$ is illustrated in fig. S2B. In the third step (fig. S2C), the planner calculates the desired takeoff velocity $\dot{\mathbf{p}}_{\mathrm{PLAN,TO}}$ on the basis of $\mathbf{p}_{\mathrm{PLAN,LD}}$ and the desired hopping height h_{D}. Assuming that each hop is ballistic, the aerial phase duration is calculated as $t_{\mathrm{air}} = \sqrt{8h_{\mathrm{D}}/g}$. The desired takeoff velocity $\dot{\mathbf{p}}_{\mathrm{PLAN,TO}}$ is calculated as

$$\dot{\mathbf{p}}_{\mathrm{PLAN,TO}} = \frac{1}{\sqrt{\frac{8h_{\mathrm{D}}}{g}}}(\mathbf{p}_{\mathrm{PLAN,LD}} - \hat{\mathbf{p}}_{\mathrm{LD}}) + \mathbf{e}_3\sqrt{2gh_{\mathrm{D}}} \quad (5)$$

where the first term represents the lateral component, and the second term represents the vertical component of the takeoff velocity.

The modeling, estimation, and planning processes in Eqs. 1 to 5 give the planned takeoff velocity $\dot{\mathbf{p}}_{\mathrm{PLAN,TO}}$ at the start of the next hopping cycle. On the basis of the leg drop experiments, we can relate takeoff velocity and landing angle through the empirically fitted equations

$$||\dot{\mathbf{p}}_{\mathrm{TO}}|| = k_v||\dot{\mathbf{p}}_{\mathrm{LD}}|| \quad (6)$$

$$\psi = k_\theta\theta_{\mathrm{LD}} \quad (7)$$

Here, Eqs. 6 and 7 predict the takeoff speed and angle, respectively. To minimize the control effort, our design does not actively control the robot landing velocity. In the fourth step (fig. S2D), the planner calculates the target landing attitude $\mathbf{z}_{\mathbf{b}_{\mathrm{D}}}$ on the basis of the empirical mappings and the desired takeoff velocity $\dot{\mathbf{p}}_{\mathrm{PLAN,TO}}$. In comparison to a prior hopping design (*20*), this empirical planning approach is computationally efficient—suitable for high-frequency and low-height hopping. The controller does not analytically plan for the robot takeoff attitude, and the robot stance phase is completely passive. This passive and computationally efficient design also improves hopping adaptability. During the passive stance phase, the elastic forces dominate gravitational forces. As the elastic force aligns with the leg axis regardless of ground inclination (Fig. 1E), the stance phase dynamics remain largely unaffected by surface tilt. This feature enables the robot to hop on inclined surfaces without requiring modifications to the dynamic model or control methods.

In the fifth step (fig. S2E), a proportional-derivative attitude controller controls the robot heading $\mathbf{z}_{\mathbf{b}}$ during robot descent. The robot impacts the ground passively and quickly bounces upward. In the final step (fig. S2F), an altitude controller estimates powered ascent duration and injects energy by commanding a thrust force. The thrust force is switched off after the robot gains sufficient energy to reach the desired jump height. This six-step process repeats for every hopping cycle.

Statistical analysis

The boxplots in Fig. 2E illustrated lateral position error distribution of four trajectory patterns ($n = 5$ for 4-, 7-, and 10-cm hopping, and $n = 3$ for 20-cm hopping). The boxplots in Fig. 3 (D and F) illustrate the commanded thrust, torques, and estimated power corresponding to the same flights shown in Fig. 2E. The boxplots in Fig. 4 (B and C) illustrated commanded thrust and torque for hopping on different terrains ($n = 5$ for soil, ice, glass, wood, and grass). The boxes showed 25, 50, and 75% percentiles, and the black bars showed minimum and maximum position errors. The data values are presented in table S2 and S3.

Supplementary Materials

The PDF file includes:

Supplementary Discussion S1 to S3
Figs. S1 to S21
Tables S1 to S3
Legends for movies S1 to S7

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S7

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